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A data-driven model simulating primary infection probabilities of cucumber downy mildew for use in early warning systems in solar greenhouses

Chun-Jiang Zhao[a,b,*], Ming Li[a,b], Xin-Ting Yang[a,b], Chuan-Heng Sun[a,b], Jian-Ping Qian[a,b], Zeng-Tao Ji[a,b]

[a] National Engineering Research Center for Information Technology in Agriculture, Beijing 100097, PR China
[b] Key Laboratory for Information Technology in Agriculture, Ministry of Agriculture, Beijing 100097, PR China

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ABSTRACT

Treatment during the primary infection phase is essential for controlling cucumber downy mildew in solar greenhouses. An early warning model applicable to this phase would represent a foundation for early warning systems for managing the disease and reducing pesticide usage. Based on the input parameters that were both readily available and appropriately limited in number, EWMPICDW (early warning model for primary infection of cucumber downy mildew in solar greenhouses) was developed based on monitoring data, early warning theory and plant disease epidemiology. The elaboration of this model included clarification of the meaning of warning, monitoring the warning indicators, forecasting the warning situation, tracing the warning sources and controlling the warning situation. The definition of warning included disease occurrence (yes or no) and probability. Because the leaf wetness duration (LWD) played an important role in disease warning systems for crops in solar greenhouses and was difficult to monitor, the leaf wetness sensor and RH threshold model were investigated and combined to form a practical estimation solution for LWD. Within the warning situation forecasting model system, the infection condition and incubation early warning submodels received the most attention. The infection condition early warning submodel was developed by using a threshold method based on the combination of LWD and mean temperature in LWD. The temperature was chosen as the warning indicator for incubation, and the incubation early warning submodel was defined using nonlinear regression methods. The warning sources traceability algorithm was developed in relation to expert knowledge and in terms of a mode of disaster mitigation that involved cutting the disaster chain from the headstream. The method for controlling the warning situation was based on good agricultural practices (GAP). The early warning model was implemented as a system and was evaluated using data for 4 years at two sites in Beijing, China. The warnings can be provided more than 2 d before symptoms appear. Using EWMPICDW, a positive early warning is associated with a change in the chance of disease occurrence from 0.68 to 0.96. Accordingly, the probability of disease occurrence calculated for the early warning model was 96%. These results demonstrate that the data-driven model will support the development of early warning systems for primary infection by cucumber downy mildew in solar greenhouses.

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1. Introduction

Cucumber downy mildew, caused by *Pseudoperonospora cubensis* (Berk. & Curt.) Rostov. (Shetty et al., 2002; Yang et al., 2007), is a major problem in solar greenhouse cucumber production. The midtemperature and high-humidity environment in unheated greenhouses is suitable for the development of this disease and can result in infections and epidemics. Because the essential characteristics of the disease include a potentially high rate of infection and a high epidemic rate (Hong et al., 1989), the disease cannot be controlled at an economically suitable threshold level when symptoms appear in a solar greenhouse environment. The means to obtain early warnings of primary infection are not currently available to most growers. The control of cucumber downy mildew usually depends on repeated fungicide sprays every 7–14 d or on frequent application of fungicides when the symptoms appear. An early warning model and a related information system are needed to respond adequately to society concerns about food quality safety, environmental protection and production cost reduction.

Infection of cucumbers by *P. cubensis* has been studied internationally for many years (Arauz et al., 2010; Cohen, 1977; Lindenthal et al., 2005; Colucci, 2008). The infection forecast models belong to two general categories: empirical models (Li, 2006; Yang et al.,

* Corresponding author at: China National Engineering Research Center for Information Technology in Agriculture (NERCITA), Building A, Beijing Nongke Masion, 11# Shuguang Huayuan Middle Road, Haidian District, Beijing 100097, PR China. Tel.: +86 10 51503411; fax: +86 10 51503750.
E-mail address: chunjiangzhao@nercita.org.cn (C.-J. Zhao).

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2007) and fundamental models (Hong, 1987; Hong et al., 1989, 1990). Empirical models need more scientific and sound interpretation, whereas fundamental models are difficult to apply because of their complex parameters. The ability to integrate the two kinds of models would be potentially useful. In order to reach this goal, consideration of the dominant factor, leaf wetness duration (LWD), is essential. Because the LWD is difficult to monitor and estimate in greenhouses in the presence of interactions between microclimate and crops, most empirical models rely on the combination of temperature and humidity. However, most fundamental models include LWD as a parameter, and the value of LWD is often estimated by using the continuous hours of misting the leaves with sprayers or inoculation devices. A practical method of monitoring and estimating LWD is needed in order to develop an approach that combines the ease of measurement of parameters in empirical models with the strong mechanistic bases of fundamental models.

In the case of cucumber downy mildew, the minimal LWD for infection is 2 h. Although many studies of LWD have been performed on field crops and fruit trees (Gleason et al., 2008; Kim et al., 2006; Magarey et al., 2006a,b; Orlandinia et al., 2008; Portraz et al., 1994; Schmitz and Grant, 2009; Sentelhas et al., 2008), LWD monitoring and estimation methods for use in greenhouse crop disease warning systems are rarely reported. Most leaf wetness sensors now in use for LWD monitoring are of the artificial leaf variety (Getz, 1992). Because the interactions between microclimate and crops influence LWD, leaf wetness sensors should be calibrated in solar greenhouse conditions. For example, the wet/dry threshold and deployment methods should be investigated (Li et al., 2010b). In order to estimate LWD, most models have concentrated on dew formation (Zhang et al., 2002; Zheng et al., 1990), but the amount of guttation water was often neglected. A previous study found that the guttation water amount was at least half of the total amount of water on leaves in the middle of the rice canopy (Luo and Goudriaan, 1999). The authors' experience indicates that guttation water is also common on cucumber leaf margins in solar greenhouses. Therefore, guttation, as a contributor to leaf wetness, is of similar importance to dew formation for the early warning of disease.

An objective study of the performance of disease early warning models combined with probability estimations of disease occurrence will lead to a better understanding of the potential usefulness of these models (Yuen and Hughes, 2002). Bayes's theorem can be useful for examining the ways in which a disease early warning model (either in positive or negative terms) affects the probability of occurrence and for assistance in modelling and prediction (Huang et al., 2010). Likelihood ratios can be calculated from analyses of the sensitivity and specificity of the early warning system. These ratios provide convenient summaries of the model's performance when interpreted in terms of a simpler form of Bayes's theorem. With these methods, the disease early warning system will furnish an operational model for forecasting occurrence date and probability, which is analogous to methods used for predicting the probability of precipitation (Kumar et al., 1999; Maini et al., 2004) and of earthquakes (Papoulia et al., 2001; Tsapanos et al., 2001), etc. Although there have been some reports addressing the possibility of disease occurrence forecasting and warning systems for field crops (Mila and Carriquiry, 2004; Vincelli and Lorbeer, 1988), little work has been done on early warning systems that involve estimates of the probability of vegetable disease occurrence in solar greenhouses.

The goal of the research reported in this paper is to obtain simple, real-time and correct results from a probability-based early warning model for the purpose of system implementation. In pursuit of this goal, this paper describes a solution to the problem of LWD input that uses leaf wetness sensors and an estimation model. The study then investigated a Bayesian analysis method for an early warning model for primary infection of cucumber downy mildew in solar greenhouses (EWMPICDW). The model was developed and implemented as a system and evaluated in solar greenhouses in Beijing, China.

2. Materials and methods

2.1. Cucumber cultivation

This study was conducted using four cultivation seasons and eight solar greenhouses from 2006 to 2009. The disease-resistant cucumber (*Cucumis sativus* L.) "Jingyanmini No. 2" (Zhang et al., 2006) was used in all experiments. The first experiment in the autumn–winter season was carried out at Beijing Xiaotangshan Precision Agriculture Experimental Base (Xiaotangshan), in Changping district, Beijing, PR China (40.18°N, 116.47°E) from October 2006 to January 2007. Three greenhouses were used in the experiment. Each greenhouse was 30 m × 7 m and was constructed of metal arches covered with polyethylene films. The row of three greenhouses was oriented in a north–south direction. The greenhouses were not heated, and they had side vents that were opened for 5–8 h daily when the weather permitted. The cucumber seedlings were planted on October 1st and were transplanted on October 13, 2006, and the harvest season ended on January 23, 2007. Five hundred and four plants were arranged in a double-row pattern with small spaces of 40 cm and large spaces of 110 cm between rows, and there was 40 cm between plants in the rows. The substrate used was a 2:1 mixture of peat and vermiculite with 500 g chicken manure and 10 g N15-P10-K15 compound fertiliser per plant. The cucumbers were maintained according to the recommendations of the Beijing Vegetable Research Center (BVRC), China, and no fungicides were applied to treat *P. cubensis* until the time of first disease onset. The second experiment in the autumn–winter season was conducted at Beijing Academy of Agriculture and Forestry Sciences (BAAFS), in Haidian district, Beijing, PR China (39.95°N, 116.29°E) from September 2008 to January 2009. The greenhouse (5 m × 4 m) was one of six rooms in a large greenhouse (north–south orientation). The greenhouse was constructed of metal arches covered with polycarbonate sheeting. A brick wall formed the eastern end of the greenhouse, and the western end was a glass window. The greenhouse used a "damp curtain-ventilator" for both vertical ventilation and cooling. The cucumber seedlings were planted on September 28th and transplanted on October 12, 2008, and the harvest season ended on January 19, 2009. Sixty-nine plants were arranged in the double-row pattern with small spaces of 40 cm and large spaces of 100 cm between rows, and there was 40 cm between plants in the rows. The substrate and cultivation practices were similar to those used in the first experiment.

The first experiment in the spring–summer season was performed at Xiaotangshan from January 2007 to August 2007. The cucumber seedlings were planted on January 7th and transplanted on February 5th. The harvest season ended on August 21, 2007. The second experiment in the spring–summer season was performed at BAAFS from January to December 2009. The cucumber seedlings were planted on January 7th and transplanted on February 5th. The harvest season ended on December 28, 2009. The greenhouses and substrate used in the spring–summer season were the same to those used in the autumn–winter season.

2.2. Microclimatic information collection

2.2.1. Xiaotangshan

Beginning on the transplanting date, the air temperature and relative humidity in the greenhouse were recorded every 30 min by sensors coupled to a datalogger (Portable equipment for measuring

temperature, relative humidity (RH) and dew point temperature, National Engineering Research Center for Information Technology in Agriculture, Beijing, China). Five sensors were arranged in an "X" pattern in each greenhouse, and 15 total sensors were used. The sensors were located in the centre of each group of crops in the greenhouse and shielded from direct solar radiation by suitable shelters. At the beginning of cucumber growth, the sensors were 10 cm higher than the plants. The sensors were moved upwards regularly as the plants grew. When the plants reached the height of the iron wires used for hanging the stems, the sensors were fixed within the central part of the canopies. The outside weather parameters, temperature, relative humidity and solar radiation were also recorded by an outside weather station every day at 30 min intervals. The data were transferred from the datalogger (RS485 connection) to a central computer (RS232 connection) via a local area network (LAN) and stored in a Microsoft Access 2003 database.

2.2.2. BAAFS

Beginning on the transplanting date, the same sensors used in Xiaotangshan were used in the experimental greenhouse of BAAFS. The sensors were arranged above rows No. 1, 2, 3, 5 and 6. They were located in the southern, middle and northern part of the greenhouse. The deployment and usage of sensors was similar to Xiaotangshan.

2.3. Disease investigation

The inocula were the pathogens that naturally occurred in these greenhouses. Each investigation was conducted by the same person in the morning before the leaves were dry. After the seedlings were transplanted, an extensive survey was carried out each day until greenish infected tissues appeared. The date of the first occurrence was recorded. In Xiaotangshan, five groups of 15 plants in an "X"-pattern around the location of sensors were selected for disease observation in each greenhouse. These groups were named South-west, North-west, Middle, South-east and North-east. In Xiaotangshan, the total number of points for observation was 15. In BAAFS, 15 groups of four plants in a parallel-line pattern around the sensor locations were selected for use as points for observation to provide 15 total as in Xiaotangshan. In all, the investigation included $15 \times 4 = 60$ points for observation over the four cultivation seasons. These points were numbered from 1 to 60. In this paper, we summarised our investigations from the earliest predicted infection date to the latest observed disease occurrence date for the 60 points. The resulting sample included 735 cases.

3. Model descriptions

3.1. Conceptual model

It is obvious that the early warning models will be suitable in practice only when input parameters are readily available and appropriately limited in number (Li et al., 2010a). EWMPICDW was developed based on early warning theory and plant disease epidemiology, which included clarification of the meaning of warning, monitoring the warning indicators, forecasting the warning situation, tracing the warning sources and controlling the warning situation (Fig. 1).

3.2. Clarifying the meaning of warning

The goal of this component is to confirm the object of the warning. The object addressed in the paper is the cucumber downy mildew. The indicators of measured disease include disease occurrence probability, disease incidence, disease index, etc. Because

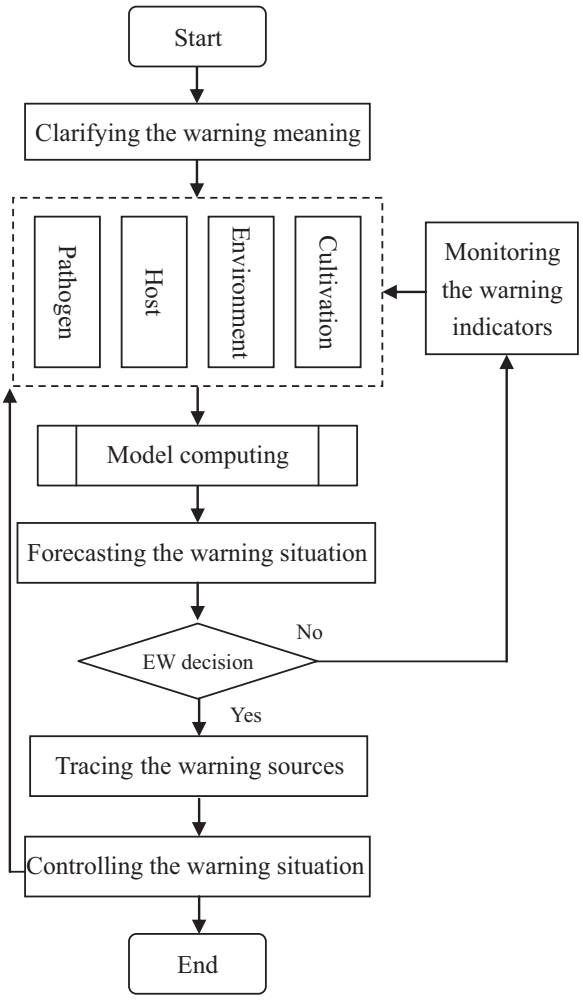


Fig. 1. The conceptual model of the early warning model for primary infection of cucumber downy mildew in solar greenhouses (EWMPICDW).

the cucumber downy mildew spreads rapidly under favourable weather conditions, growers are primarily concerned about the possibility that infection will occur, rather than about knowing how much infection will actually take place (Arauz et al., 2010). Accordingly, the probability of disease occurrence was selected as the object of the warning for the purposes of this paper. EWMPICDW should provide early warnings before the symptom appears. These warnings would include disease occurrence (yes or no) and its probability. These warnings would be beneficial because they could serve to reduce the frequency of fungicide sprays and likewise to improve the quality of the agricultural products.

3.3. Monitoring the warning indicators

Among the warning indicators including temperature, humidity, etc., the LWD receives the most attention. Because the leaf wetness duration (LWD) is difficult to monitor and is a key input to disease warning systems for crops in solar greenhouses, the leaf wetness sensor and RH threshold model were combined to form a practical estimation solution for LWD. In our experiment, the LWD was mainly due to guttation.

3.3.1. Leaf wetness sensor calibration

Five leaf wetness sensors (#6420, Davis Instruments Corp., CA, USA) and a datalogger (CR1000, Campbell Scientific, Logan, UT, USA) were used for LWD monitoring. The calibration method for the



Fig. 2. The deployment of leaf wetness sensors in solar greenhouse for cucumbers.

leaf wetness sensors was developed using minicucumbers at the fruit harvesting stage in the solar greenhouse. The leaf wetness sensors were oriented at a 45° angle. They were first misted with water from a pipette and then attached to the wet leaves. The dry/wetness threshold was determined under these two different conditions. The monitoring results from different sensor deployments were compared. These deployments included attaching sensors to the leaf margins, tips and lower surface and placing them below the leaf (Fig. 2). The monitoring results yielded a value of 6 for the dry/wetness threshold when the sensors were attached to the wet leaves. The monitoring effects were considered good, and the errors were found to be around 1 h in this case. When the sensors were attached to the leaf margins and tips, the monitoring accuracy (fraction of estimates judged to be correct) was around 0.75–0.83. These results indicate that the calibration method for leaf wetness duration could meet the requirements of the early warning system (Li et al., 2010b).

3.3.2. Leaf wetness duration estimation model based on canopy relative humidity

In continuous cloudy weather, the cucumber leaf wetness duration is longer, and the relative humidity in solar greenhouses is higher. The LWD estimation model based on canopy relative humidity (RH threshold model) is one of the easiest models to implement and to use (Sentelhas et al., 2008), but it should be calibrated and validated under real greenhouse conditions. Note that

$$\mathrm{RH} \geq TR \quad (0 \leq TR \leq 100\%) \tag{1}$$

where RH is relative humidity (%), and TR is the relative humidity threshold at leaf wetness (%).

The RH data were obtained at 5 min intervals from an automatic weather station (Vantage Pro & Plus, Davis Instruments Corp., CA, USA) located in the middle section of the greenhouse. The trial-and-error method (Xiong, 2009), average value method (Gillespie et al., 1993) and the method based on frequency of leaf wetness (Zheng et al., 1990) were used to calibrate the RH threshold. The RH thresholds obtained using these three methods were 90%, 89% and 93%, respectively. To illustrate our approach using the trial-and-error method, we developed a calibration program using Visual C# 2005 and Microsoft Access 2003. The algorithm is follows:

(1) Create the data tables for observed values of leaf wetness duration and relative humidity data in greenhouses at the mean time.
(2) Determine the primary values of TR as 80%.
(3) The estimated values of leaf wetness duration are computed using the model RH ≥ TR, and TR increases from 80% to 100% by 1% intervals.
(4) The root mean square error ($RMSE$) of the differences between estimated values and observed values is used as a criterion for the model performance. The optimal parameter value of TR = 90% is obtained by minimising $RMSE$.

The trial-and-error approach (MAE = 2.03) and average value approach (MAE = 2.14) were superior to the method based on frequency of leaf wetness. The errors were around 2 h. Because the LWD values were over 3 h, the monitoring effects of the RH threshold (RH = 89% or 90%) model were considered acceptable (Li et al., 2010c).

3.4. Forecasting the warning situation

Based on the mechanisms of cucumber downy mildew infestation (Lindenthal et al., 2005), the infection condition and incubation period early warning submodels were seen to represent key stages for primary infection and were given more attention (Fig. 3). The interval between inoculation and the appearance of disease symptoms is known as the incubation period (Agrios, 2005). Disease symptoms appear following the incubation period.

3.4.1. Infection condition early warning submodel

Data from a literature on experimental infection of susceptible cucumber leaves by *P. cubensis* under controlled conditions (Cohen, 1977) were used for modelling infection. These data indicate that the minimal temperature for infection is 20°C with 2 h of wetness (Cohen, 1977). The infection condition therefore depends on a favourable combination of LWD and mean temperature in LWD. The minimal threshold is 2 h × 20 °C = 40 h °C. This threshold agrees with many mechanistic experimental results (Hong, 1987; Shi et al., 2005). For the recovery period subsequent to transplanting cucumber seedlings, the model would run beginning on a date 1 week after transplanting. Note also that

$$\mathrm{LWD} \times \mathrm{TLWD} \geq 40\,\mathrm{h}\,^{\circ}\mathrm{C} \quad (\mathrm{LWD} \leq 2\mathrm{h},\ 5\,^{\circ}\mathrm{C} \leq \mathrm{TLWD} \leq 30\,^{\circ}\mathrm{C}) \tag{2}$$

where LWD is leaf wetness duration (h), and TLWD is mean temperature (°C) in leaf wetness duration.

3.4.2. Incubation period early warning submodel

The time of first appearance of downy mildew symptoms depends more on temperature than on leaf wetness duration (Yang et al., 2007). Published data from inoculation experiments at different temperatures suggest the following values for the elapsed time between infection and symptom appearance: 2 d at 22–25 °C, 3–4 d at 26–30 °C, 7 d at 15–16 °C and 10 d at 12–15 °C (Fu and Yao, 1983). The temperature contribution rate (y) for symptom appearance was therefore measured by the reciprocal of the duration (d) of the inoculation period in the controlled experiments. The relationship of hourly mean temperature (t) and temperature contribution rate (y) was fitted using the SAS (Statistical Analysis System V.8.0.2, SAS Institute Inc., Cary, NC, USA) NLIN program with the Gauss–Newton method (Fig. 4) and expressed as:

$$y = \frac{0.0165}{1 + 10389.2 \times \exp(-0.5743 \times t)} \quad (p = 0.0033) \tag{3}$$

where y is contribution rate, and t is hourly mean temperature (°C).When y attains a value of 1, the incubation period is over and symptoms may appear.

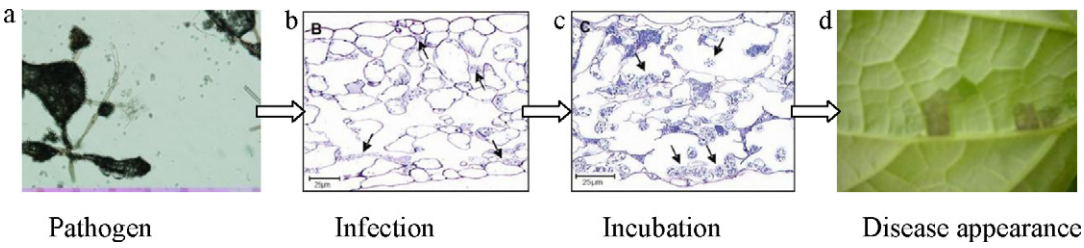


Fig. 3. The primary infection mechanism of cucumber downy mildew.

3.5. Tracing the warning sources

Successful tracing of a disease warning to its source provides the basis of controlling disease for solar greenhouse vegetables. Using disaster chain-style theory and plant epidemiological theory (Xiao, 2006), the problem of tracing a disease warning to its source was systematically analysed in association with disaster mitigation modes that involved cutting the disaster chain from the origin. The model took into account the multiple sources of disease early warning under solar greenhouse conditions and their effects on the modes and paths of interpretation of disease early warning. For example, because *P. cubensis* is a widely existing pathogen in cucumber production fields, and because greenhouses allow cultivation of cucurbits throughout the year, the primary inoculum sources exist in most greenhouses (Yang et al., 2007) as the warning source. In the winter and spring, overcast, rainy, snowy and foggy weather conditions also represent potential sources of warnings of cucumber downy mildew infection. To integrate this experience and knowledge, the warning source traceability algorithm for cucumber downy mildew in solar greenhouses was constructed and tested for ease of system realisation. Detailed information about this project appears in our publication (Li et al., 2010d).

3.6. Controlling the warning situation

Based on good agricultural practices (GAP), the control of the warning situation was established using the method of mitigation disaster by cutting the disaster chain upstream from the focus of the problem. For example, the overcast, rainy, snowy and foggy weather often represents the warning source for the occurrence of cucumber downy mildew. Farmers can enhance ventilation so as to decrease the LWD and can thereby control the conditions that would otherwise lead to infection. If the user chooses to apply chemicals, then the agents used should be suitable and justified (according to label recommendations or official registration body publication) for *P. cubensis* based on CHINAGAP (Yang et al., 2007).

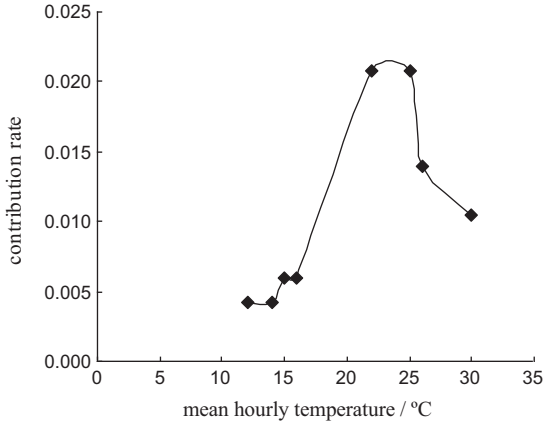


Fig. 4. Fitted curve of the mean hourly temperature and contribution rate for incubation of cucumber downy mildew in solar greenhouses.

3.7. Model evaluation method

Analyses of sampling variance and standard deviation were performed using SAS8.02. The figures were drawn using Microsoft Excel 2003. The main methods used are listed below.

3.7.1. Error analysis method

The estimated dates and observed dates of primary infection of cucumber downy mildew were compared using random point figures. Also we concluded some statistical indicators to estimate the errors of early warning for disease occurrence. The errors were evaluated by *RMSE*, Willmott agreement index (*W*), mean absolute error (*MAE*) and mean bias error (*MBE*) (Sentelhas et al., 2008).

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(Pi - Oi)^2}{N-1}} \quad (4)$$

$$W = 1 - \frac{\sum_{i=1}^{N}(Pi - Oi)^2}{\sum_{i=1}^{N}(|P_i'| + |O_i'|)} \quad (5)$$

$$MAE = \frac{\sum_{i=1}^{N}|P_i - O_i|}{N} \quad (6)$$

$$MBE = \frac{\sum_{i=1}^{N}(P_i - O_i)}{N} \quad (7)$$

where P_i are estimated values, O_i are observed values, P_i' are deviations from average estimated values, O_i' are deviations from average observed values, and N is sample size.

3.7.2. Bayesian analysis method

To evaluate the 735 cases of investigation and compute the probability of early warning for cucumber downy mildew, the Bayesian analysis method was used and described in detail (Yuen and Hughes, 2002). If the model correctly estimated disease presence in an interval of time (the interval of time in the paper was 1 d, i.e., one time of investigation for one point), it was scored in box *X*. If the model failed to estimate disease when it did occur, it was scored in box *Y*. If the model estimated disease when it did not occur, it was scored in box *S*. Finally, if the model correctly estimated absence of disease, it was scored in box *Z* (Table 1). The sensitivity (S_{en}), specificity (S_{pe}), accuracy (A_{cc}, i.e., Youden index), false negative rate (F_{nr})

Table 1
Categories and a summary of calculation results of the early warning model for primary infection of cucumber downy mildew in solar greenhouses (EWMPICDW) in each day.

N	Estimated – Yes	Estimated – No
Observed – Yes	Hits (*X*)	Misses (*Y*)
Observed – No	False alarms (*S*)	Correct negatives (*Z*)

Table 2
Disease investigation results in Xiaotangshan and BAAFS.

Year	Point	Predicted infection date	Predicted occurrence date	Observed occurrence date	Year	Point	Predicted infection date	Predicted occurrence date	Observed occurrence date
Xiaotangshan					BAAFS				
2006	1[a]	21-October	26-October	26-October	2008	31[b]	N	N	N
2006	2[a]	21-October	26-October	26-October	2008	32[b]	N	N	N
2006	3[a]	21-October	26-October	26-October	2008	33[b]	N	N	N
2006	4[a]	21-October	26-October	26-October	2008	34[a]	30-October	4-November	21-November
2006	5[a]	21-October	26-October	26-October	2008	35[c]	2-December	10-December	20-November
2006	6[a]	21-October	26-October	26-October	2008	36[a]	13-November	17-November	20-November
2006	7[a]	21-October	26-October	26-October	2008	37[d]	12-November	17-November	16-November
2006	8[a]	21-October	26-October	26-October	2008	38[c]	2-December	9-December	16-November
2006	9[a]	21-October	26-October	26-October	2008	39[c]	25-November	1-December	14-November
2006	10[a]	21-October	26-October	26-October	2008	40[a]	11-November	16-November	16-November
2006	11[a]	21-October	26-October	26-October	2008	41[d]	12-November	17-November	15-November
2006	12[a]	21-October	26-October	26-October	2008	42[c]	N	N	14-November
2006	13[a]	21-October	26-October	27-October	2008	43[b]	N	N	N
2006	14[a]	21-October	26-October	26-October	2008	44[d]	11-November	16-November	14-November
2006	15[a]	21-October	26-October	26-October	2008	45[d]	11-November	16-November	15-November
2007	16[a]	13-February	21-February	24-February	2009	46[d]	16-April	20-April	19-April
2007	17[d]	21-February	7-March	24-February	2009	47[b]	N	N	N
2007	18[a]	16-February	24-February	24-February	2009	48[b]	N	N	N
2007	19[a]	15-February	24-February	26-February	2009	49[d]	16-April	20-April	19-April
2007	20[d]	16-February	26-February	24-February	2009	50[b]	N	N	N
2007	21[a]	16-February	24-February	24-February	2009	51[c]	N	N	20-April
2007	22[a]	14-February	21-February	24-February	2009	52[b]	N	N	N
2007	23[a]	16-February	24-February	26-February	2009	53[b]	N	N	N
2007	24[a]	13-February	21-February	24-February	2009	54[c]	N	N	20-April
2007	25[a]	16-February	23-February	24-February	2009	55[c]	N	N	20-April
2007	26[b]	N[e]	N	N	2009	56[b]	N	N	N
2007	27[b]	N	N	N	2009	57[b]	N	N	N
2007	28[b]	N	N	N	2009	58[b]	N	N	N
2007	29[b]	N	N	N	2009	59[b]	N	N	N
2007	30[b]	N	N	N	2009	60[b]	N	N	N

[a] The predicted infection date or predicted occurrence date was earlier than the observed occurrence date.
[b] The cucumber downy mildew did not appear, and the model did not present early warning (true negative points, TNP).
[c] The model did not present infection and occurrence early warning before the observed occurrence date.
[d] The predicted infection date was earlier than the observed occurrence date, but predicted occurrence date was later than the observed occurrence date.
[e] N denoted no predicted infection, predicted disease occurrence or observed disease occurrence.

and false positive rate (F_{pr}) were then computed.

$$Sen = \frac{X}{(X+Y)} \tag{8}$$

$$Spe = \frac{Z}{(Z+S)} \tag{9}$$

$$Acc = S_{en} + S_{pe} - 1 \tag{10}$$

$$F_{nr} = 1 - S_{en} \tag{11}$$

$$F_{pr} = 1 - S_{pe} \tag{12}$$

In general, based on Bayes's theorem, let A represent disease occurrence and $P(A)$ the prior probability of disease occurrence based on the historical data. Then the prior probability of disease nonoccurrence is $P(\bar{A})$, and $P(A) + P(\bar{A}) = 1$. Next, let B represent an early warning of the disease occurrence and $\bar{B}$ an early warning of the disease nonoccurrence. Sensitivity is the conditional probability $P(B|A)$, which represents the probability of an early warning of disease occurrence given that the disease actually occurred. Similarly, $P(\bar{B}|\bar{A}) = S_{pe}$ (specificity), $P(\bar{B}|A) = F_{nr}$ (false negative rate), and $P(B|\bar{A}) = F_{pr}$ (false positive rate). From the point of view of practical early warning for diseases, we are usually less interested in sensitivity and specificity, and more concerned with posterior probabilities (i.e., conditional probabilities) such as $P(A|B)$, the probability of disease occurrence, given an early warning of occurrence, and $P(\bar{A}|\bar{B})$, the probability of nonoccurrence of disease, given an early warning of nonoccurrence.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|\bar{A})P(\bar{A})} \tag{13}$$

$$P(\bar{A}|\bar{B}) = \frac{P(\bar{B}|\bar{A})P(\bar{A})}{P(\bar{B}|\bar{A})P(\bar{A}) + P(\bar{B}|A)P(A)} \tag{14}$$

Furthermore, the likelihood ratio of a positive early warning ($LR(A,B)$) is a useful measure that contains information from both sensitivity and specificity and that serves to summarise the results.

$$LR(A,B) = \frac{P(B|A)}{1 - P(\bar{B}|\bar{A})} \tag{15}$$

i.e., $LR(A,B) = S_{en}/(1 - S_{pe})$.

Likewise, we can also define the likelihood ratio of a negative early warning ($LR(A,\bar{B})$).

$$LR(A,\bar{B}) = \frac{1 - P(B|A)}{P(\bar{B}|\bar{A})} \tag{16}$$

i.e., $LR(A,\bar{B}) = (1 - S_{en})/S_{pe}$.

$LR(A,B)$ and $LR(A,B)$ are also useful in that they yield a simplified version of Bayes's theorem. We rewrite the prior probabilities as odds (odds(event) = P(event)/(1 − P(event))). We then obtain in place of Eq. (13):

$$odds(A|B) = odds(A) \times LR(A,B) \tag{17}$$

and similarly derive the relationship for the probability of disease after a negative early warning:

$$odds(A|\bar{B}) = odds(A) \times LR(A,\bar{B}) \tag{18}$$

From Eq. (17), we can see that as long as $LR(A,B)>1$, the effect of an early warning of disease occurrence is to increase the posterior odds of disease occurrence relative to the prior odds. From Eq.

Table 3
Error analysis of the early warning results of the early warning model for primary infection of cucumber downy mildew in solar greenhouses (EWMPICDW).

EWMPICDW	*a*	*b* (h)	R^2[a]	*W*[b]	*MAE* (h)[c]	*MBE* (h)[d]
With the TNP[e]	1.00	0.31	0.94	0.98	1.04	−0.24
Without the TNP	0.95	1.37	0.83	0.95	1.62	−0.38

[a] Regression coefficient determined by simple linear regression $Y = a + bX$, where estimated values = X, observed values = Y.

[b] $W = 1 - \dfrac{\sum_{i=1}^{N}(Pi-Oi)^2}{\sum_{i=1}^{N}(|P_i'|+|O_i'|)^2}$.

[c] $MAE = \dfrac{\sum_{i=1}^{N}|P_i-O_i|}{N}$.

[d] $MBE = \dfrac{\sum_{i=1}^{N}(P_i-O_i)}{N}$ where P_i are estimated values, O_i are observed values, P_i' are deviations from average estimated values, O_i' are deviations from average observed values, and N is sample size.

[e] TNP (true negative points). In the points, the cucumber downy mildew did not appear, and the model did not present early warning.

Table 4
Bayesian analysis of early warning results of the early warning model for primary infection of cucumber downy mildew in solar greenhouses (EWMPICDW).

	Estimated – Yes	Estimated – No	Total number	Prior probability	Posterior probability
Observed – Yes	$X = 84$ $P(B\|A) = 0.68$	$Y = 39$ $P(\bar{B}\|A) = 0.32$	123	$P(A) = 0.68$	$P(A\|B) = 0.96$
Observed – No	$S = 34$ $P(B\|\bar{A}) = 0.06$	$Z = 578$ $P(\bar{B}\|\bar{A}) = 0.94$	612	$P(\bar{A}) = 0.32$	$P(\bar{A}\|\bar{B}) = 0.58$
Total number	118	464	735		

(18), as long as $LR(A, \bar{B}) > 1$, the effect of an early warning of disease nonoccurrence is to decrease the posterior odds of disease occurrence relative to the prior odds. Both of these attributes are desirable in a good early warning model of disease. Ideally, therefore, a good early warning model of disease has $LR(A,B)$ as large as possible and simultaneously has $LR(A, \bar{B})$ as small as possible.

4. Results

4.1. The early warning error of the model

Early warning results about disease occurrence were evaluated using the data from two sites over 4 years (Table 2). Cucumber downy mildew was found in all years but not at all observed points. In Xiaotangshan, the EWMPICDW provided good early warning results (the successful rate was $28/30 \times 100\% = 93.3\%$). This output allowed 0 to 3 d of lead time before the occurrence of disease (points 1–16, 18–19, 21–25) and obtained corrected results of disease nonoccurrence (points 26–30). Only for points 17 and 20, when the first lesion was detected before the predicted occurrence date, did the model fail to provide early warning before actual lesion appearance. However, the predicted infection date was consistently earlier than the observed occurrence date in Xiaotangshan. This information also served to alert users. In BAAFS, the EWMPICDW did not perform as well as in Xiaotangshan. The successful rate was $17/30 \times 100\% = 56.7\%$. The model provided 0 to 3 d of lead time before the occurrence of disease (points 34, 36 and 40) and yielded corrected results of disease nonoccurrence (points 31–33, 43, 47, 48, 50, 52, 53, 56–60). The predicted infection date was earlier than the observed occurrence date at points 37, 41, 44, 45, 46 and 49, which also furnished an alert to users. At points 35, 38, 39, 42, 51, 54 and 55, when the first lesion was observed before the predicted infection and occurrence date, the model failed to provide early warning before lesions appeared. The most likely source of these errors was the estimation of LWD. The RH in BAAFS was lower than in Xiaotangshan, owing presumably to urban heat island effects in BAAFS. As a result, the LWD estimates for BAAFS based on RH were lower in values or even zero. These values could not trigger an early warning of infection conditions. The model should be improved in order to reduce the number of points that involve such omissions.

The disease data between 2006 and 2009 that we investigated included 41 (of 60) points that exhibited occurrence of cucumber downy mildew (Table 2). Thus $P(A) = 41/60 = 0.68$, and $P(\bar{A}) = 0.32$. The data included 19 true negative points (TNP, i.e., Z in Table 1), i.e., points without cucumber downy mildew. These points would

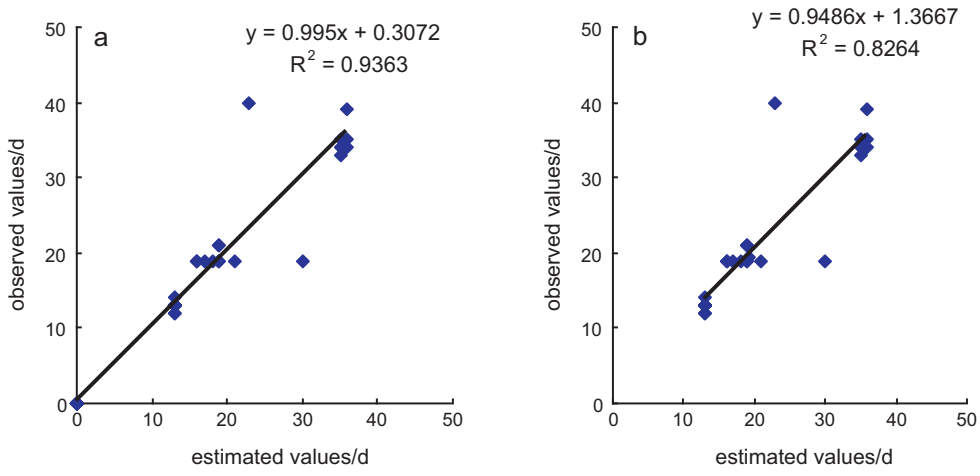


Fig. 5. Comparison of the occurrence date between observed values and estimated values under the early warning model for primary infection of cucumber downy mildew in solar greenhouses (EWMPICDW).

have large effects on the fitting results. We compared the error analyses by constructing and comparing two cases, one including the true negative points and the other not including the true negative points (Table 3). We concentrated on the results associated with points that generated early warnings of disease occurrence. The presence of TNP would decrease the *RMSE* (Fig. 5a, *RMSE* = 2.97; Fig. 5b, *RMSE* = 3.71). In the error analysis with the TNP, the model performed well, with accuracy given by $R^2 = 0.94$, $W = 0.98$. The estimation error was less than 1 d (*MBE* = −0.24). This value indicated that EWMPICDW could provide early warning no more than 1 d before the initial occurrence of disease. From the error analysis without the TNP, the fitting effects were lower than in the analysis with TNP ($R^2 = 0.83$, $W = 0.95$), but both analyses had similar estimation errors in terms of *MBE*. However, because the EWMPICDW could warning infection conditions and because the incubation period was often more than 2 d, the model could provide an early warning more than 2 d before the first appearance of symptoms (Table 2). This result meets the requirement of control practices in solar greenhouses. For example, application of the Chlorothalonil and ecological control method requires 1 d of preparation before the predicted disease occurrence date.

4.2. Early warnings of disease probability using the model

From Table 4, the model had good performance: sensitivity was 0.68, specificity was 0.94 and accuracy was 0.62. Using the EWMPICDW, a positive early warning increased the chance of disease occurrence from 0.68 to 0.96. The model would generate a warning of cucumber downy mildew occurrence with a probability of 96%. Although a negative early warning decreased the occurrence chance from 0.68 to 0.58, the probability of disease occurrence was still near 0.5; disease occurrence and nonoccurrence were equally likely, despite the negative early warning. The early warning for nonoccurrence should therefore be improved.

In the likelihood ratio form of Bayes's theorem, $LR(A,B) = 12.29 > 1$, $LR(A, \overline{B}) = 0.39 < 1$; therefore, the model yielded relatively ideal results.

4.3. Model for system development

We developed the early warning system based on the model using Visual C# 2005 and Microsoft Access 2003. The user could collect information about the cucumbers, the pathogen, cultivation and environmental conditions and generate early warnings of infection and occurrence of cucumber downy mildew (Fig. 6).

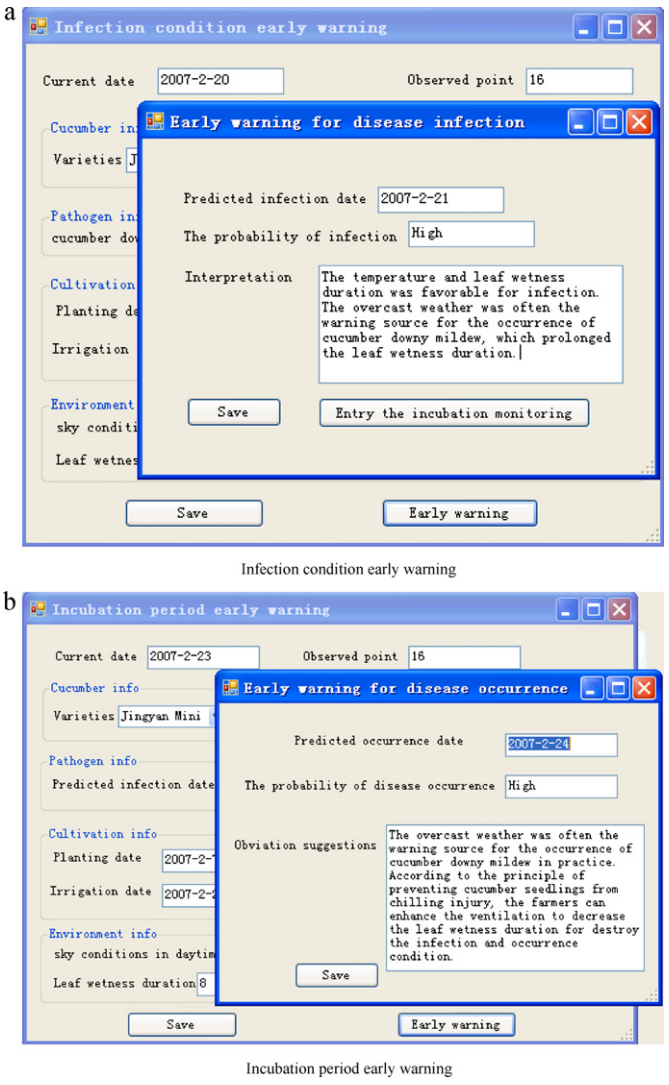


Fig. 6. Early warning system for primary infection of cucumber downy mildew in solar greenhouses.

5. Discussion

Compared with the existing model and system for early warning for cucumber downy mildew in solar greenhouses (Li et al., 2010a), the model incorporating LWD offers better mechanistic features. There are two kinds of early warning models for plant diseases (Madden and Ellis, 1988). Fundamental models have greater accuracy and scientific meaning, but they are difficult to apply because of their complex parameters (Hong, 1987; Hong et al., 1989, 1990). Empirical models are simpler, but they require additional scientific interpretation and validation (Li, 2006; Yang et al., 2007). Many previous models specified relative humidity and temperature as main inputs (Ju, 2006; Xu, 2004). However, these models paid no attention to the leaf wetness duration, which is a parameter that is essential for determining the conditions for pathogen infection in leaf diseases, e.g., in cucumber downy mildew. The model EWMPICDW integrated these two kinds of models and extracted the dominant factors. Leaf wetness duration monitoring using sensors and estimation using a relative humidity-based model were investigated in order to support the early warning model and system. A solution based on model error analysis was also provided. The model could provide early warnings more than 2 d before actual disease appearance. Based on Bayes's theorem, our approach could warn users about cucumber downy mildew occurrence with a probability of 96%. The objective study of the performance of the early warning model will lead to a better understanding of disease occurrence probability. Such understanding will prove more beneficial than the information furnished by subjective experiences. This same sort of improvement has been seen in the area of weather forecasting (Maini et al., 2004) and earth quake predicting (Papoulia et al., 2001).

Like all models, the EWMPICDW sometimes provided incorrect early warnings. The frequency of incorrect early warnings will influence the rate of system adoption by potential users. The first reason for incorrect early warnings might be the accuracy of leaf wetness duration monitoring methods and estimation models. Unlike air temperature, LWD is a difficult variable to monitor or estimate because it is driven both by atmospheric conditions and by their interactions with the structure and composition of the crop canopies (Sentelhas et al., 2005). Leaf wetness sensors may not simulate the real situation, and a model based only on relative humidity would produce some errors. Errors in LWD input, e.g., not detecting

the LWD appearance, would prevent the model from generating a disease warning. Further study of more accurate methods for LWD monitoring and estimation is perhaps needed (Li et al., 2010b).

The second reason for error would be the limited information on the prior probability of cucumber downy mildew occurrence. In reality, we often have little information on the prior occurrence of diseases. Although we estimated the prior probability of cucumber downy mildew infection using a 4-year, 2-site data set, the use of historical data in order to estimate the prior probabilities involved assumptions about similar weather, cultivars, cultural practices, and pathogen populations for other sites (Yuen and Hughes, 2002). These assumptions may or may not be true. In order to extend the model, it is essential to obtain more disease occurrence data from different years and sites for future validation.

Nevertheless, our results suggest that the LWD monitoring method and estimation model can provide LWD values sufficiently accurate to implement disease early warning systems that use LWD as a main input. Knowing the performance of EWMPICDW in probabilistic terms will enable the targeting of areas that represent good potential for use. With further testing, the data-driven model may help to encourage growers in solar greenhouses to apply disease early warning systems for reducing the frequency of fungicide sprays and thereby for improving the quality of agricultural products.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at 10.1016/j.compag.2011.02.009.

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